Validação cruzada e otimização

CROSS-VALIDATION

5 folds

```
from sklearn.model_selection import cross_val_score
s=cross_val_score(pipe,Xtr,ytr,
 cv=5,scoring="f1_macro")
print(s.mean(),s.std())
```

Pipeline evita vazamento dentro CV.

Estratificada

```
from sklearn.model_selection import StratifiedKFold
cv=StratifiedKFold(5,shuffle=True,
 random_state=42)
```

Adequado para classificação.

GRID / RANDOM

GridSearchCV

```
from sklearn.model_selection import GridSearchCV
g=GridSearchCV(pipe,param_grid,
 cv=cv,scoring="f1_macro",n_jobs=-1)
g.fit(Xtr,ytr)
```

Tuning deve excluir Xte.

RandomizedSearchCV

```
from sklearn.model_selection import RandomizedSearchCV
r=RandomizedSearchCV(pipe,dist,
 n_iter=20,cv=cv,random_state=42)
```

Busca amostral de configurações.

FINALIZAR

Parâmetros

```
print(g.best_params_)
print(g.best_score_)
```

best_score_ usa folds de treino.

Teste final

```
from sklearn.metrics import f1_score
score=f1_score(yte,g.predict(Xte),
 average="macro")
```

Não volte a otimizar com Xte.

Grupos

```
from sklearn.model_selection import GroupKFold
cv=GroupKFold(n_splits=5)
```

Passe groups quando necessário.